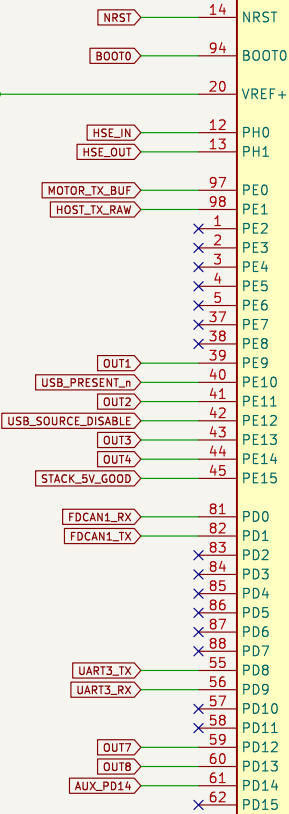
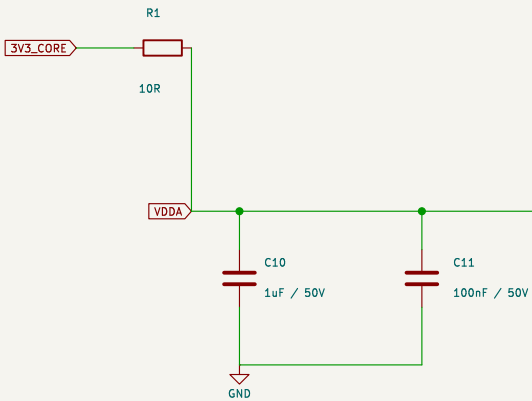
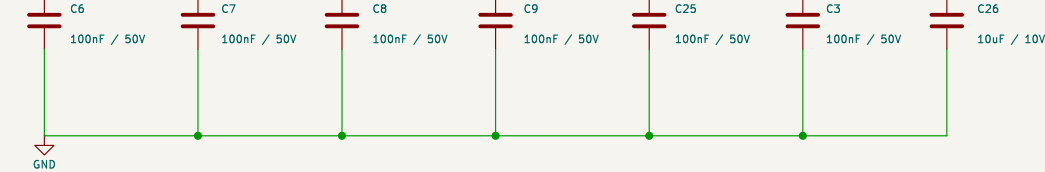


H743 CORE / LOCAL POWER NETWORKS

Filtered analog supply / VREF+



U1
STM32H743VIT6



VDD + VBAT: six 100 nF bypass capacitors and 10 uF bulk

Internal LDO mode. VCAP is not 3V3 and carries no external load.
Two local 2.2uF low-ESR capacitors; ESR <100mOhm.
AN4938 Rev7 requires VCAP pins tied. VDDA/VREF locally bypassed.
USB supply is internally bonded to VDD in this LQFP100 package.

OUT1-4 TIM1 / OUT5-6 TIM8 / OUT7-8 TIM4.
Three timer-rate groups; DMA/DShot needs bench validation.
Dedicated SPI1 IMU / SPI2 NAND / SPI3 external bus.
GPIO alternate roles share physical pins.
Raw ADC needs conditioning; no direct LiPo or servo power.

Internal LDO: VCAP48 + VCAP73 tied together
Two 2.2 uF low-ESR capacitors; no external load

power	link	pads
File: power.kicad_sch	File: peripherals.kicad_sch	File: edge-pads.kicad_sch
storage	clock	sockets
File: storage.kicad_sch	File: clock-reset.kicad_sch	File: slide-sockets.kicad_sch

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /
File: controller.kicad_sch

Title: H743 CORE / BYPASS / VCAP

Size: A3
KiCad E.D.A. 9.0.2

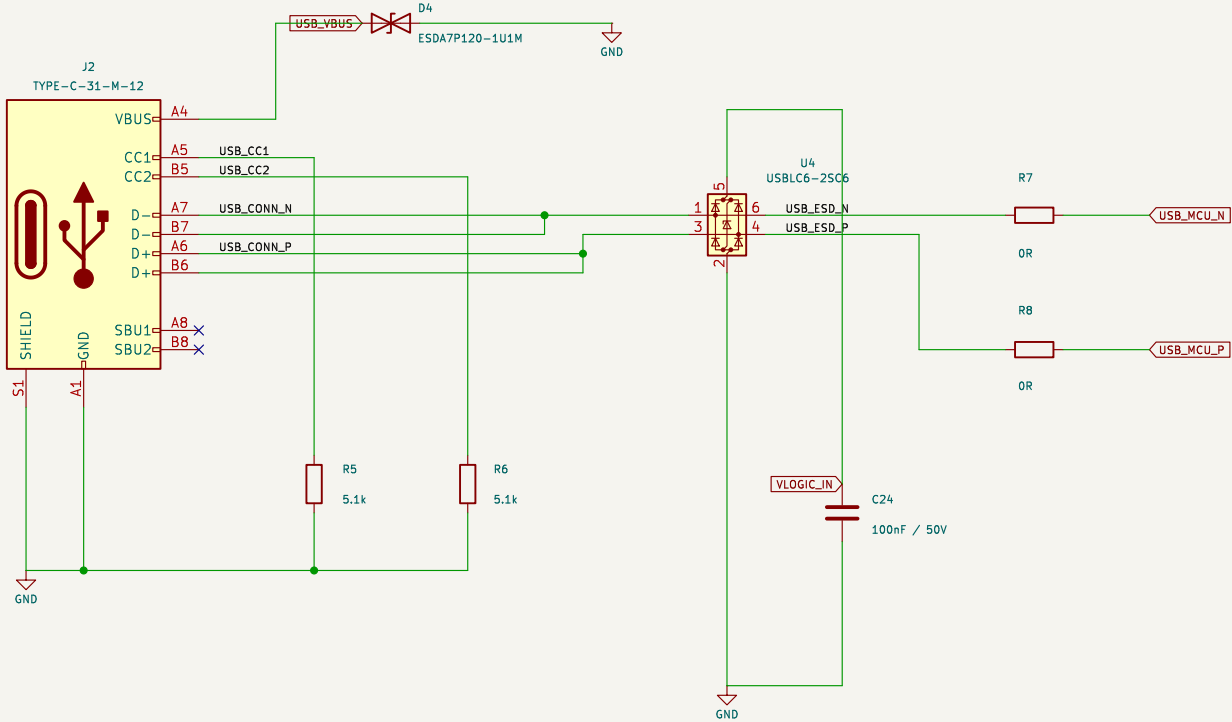
Date: 2026-10-03

Rev: R3-S6-P6

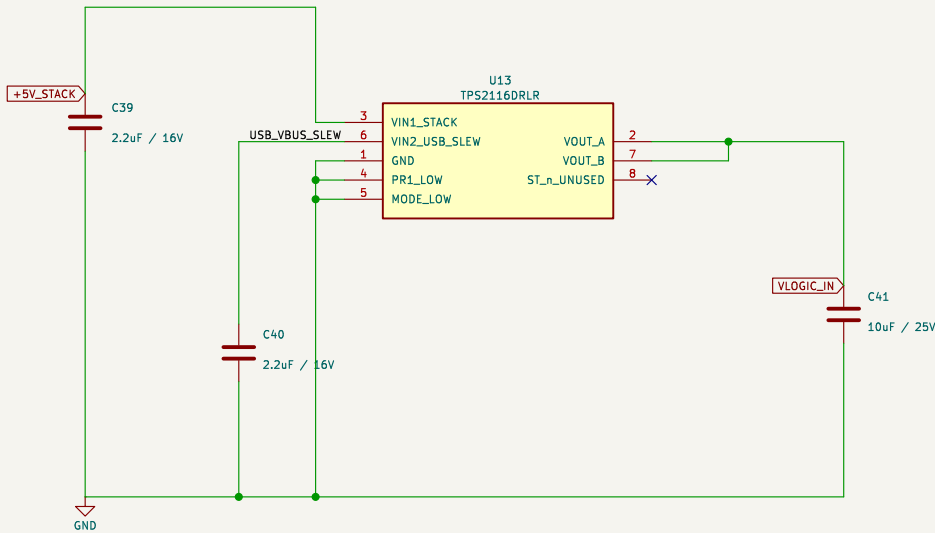
Id: 1/7

USB INPUT / SOURCE SELECTION / 3.3 V POWER

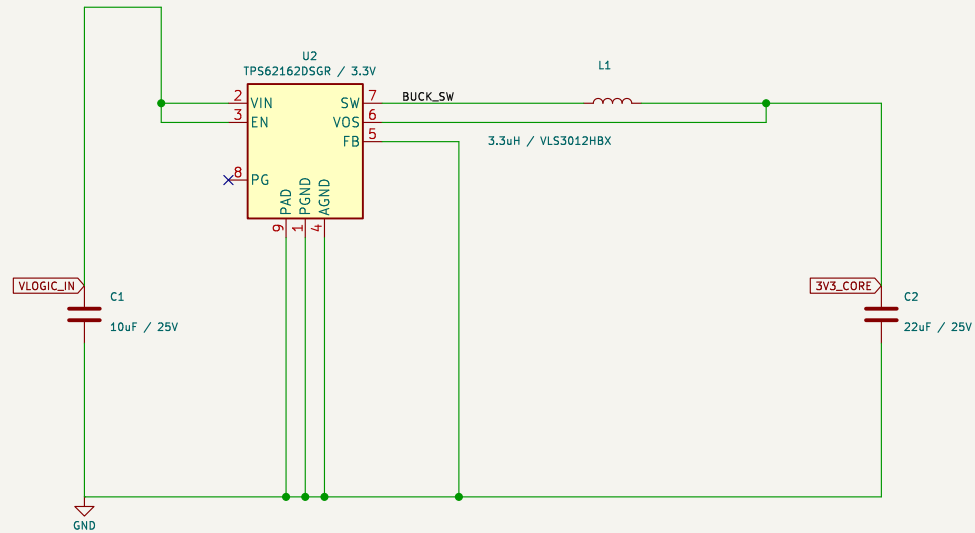
USB-C termination and ESD



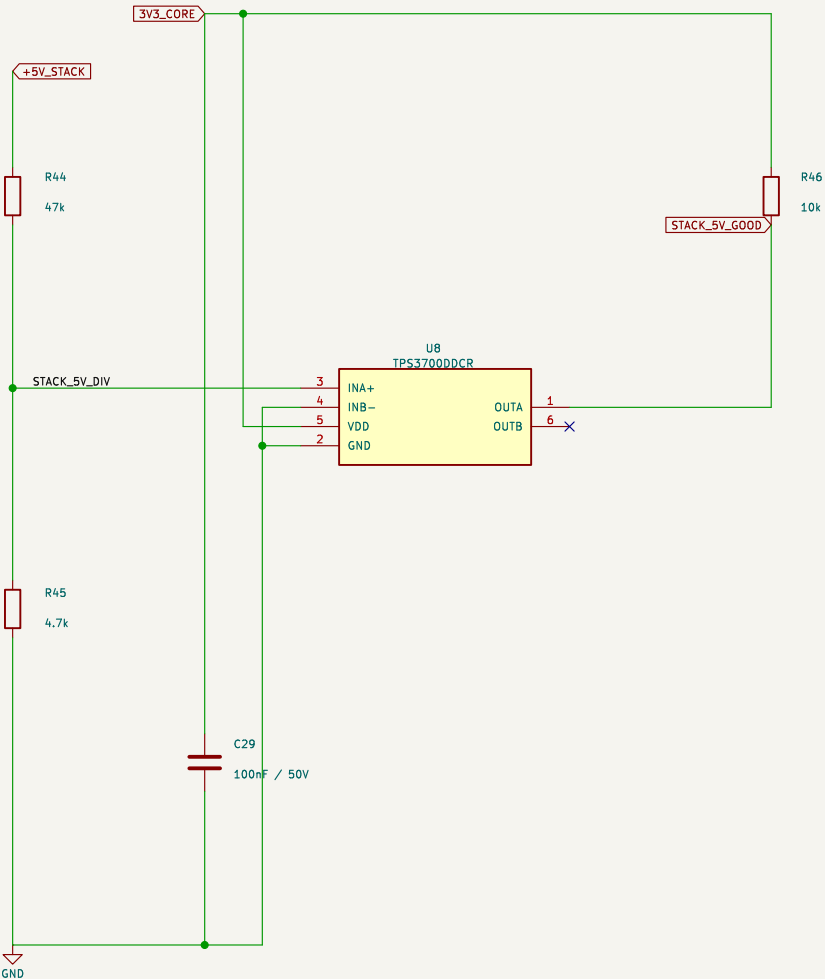
Power mux: stack / slew-limited USB



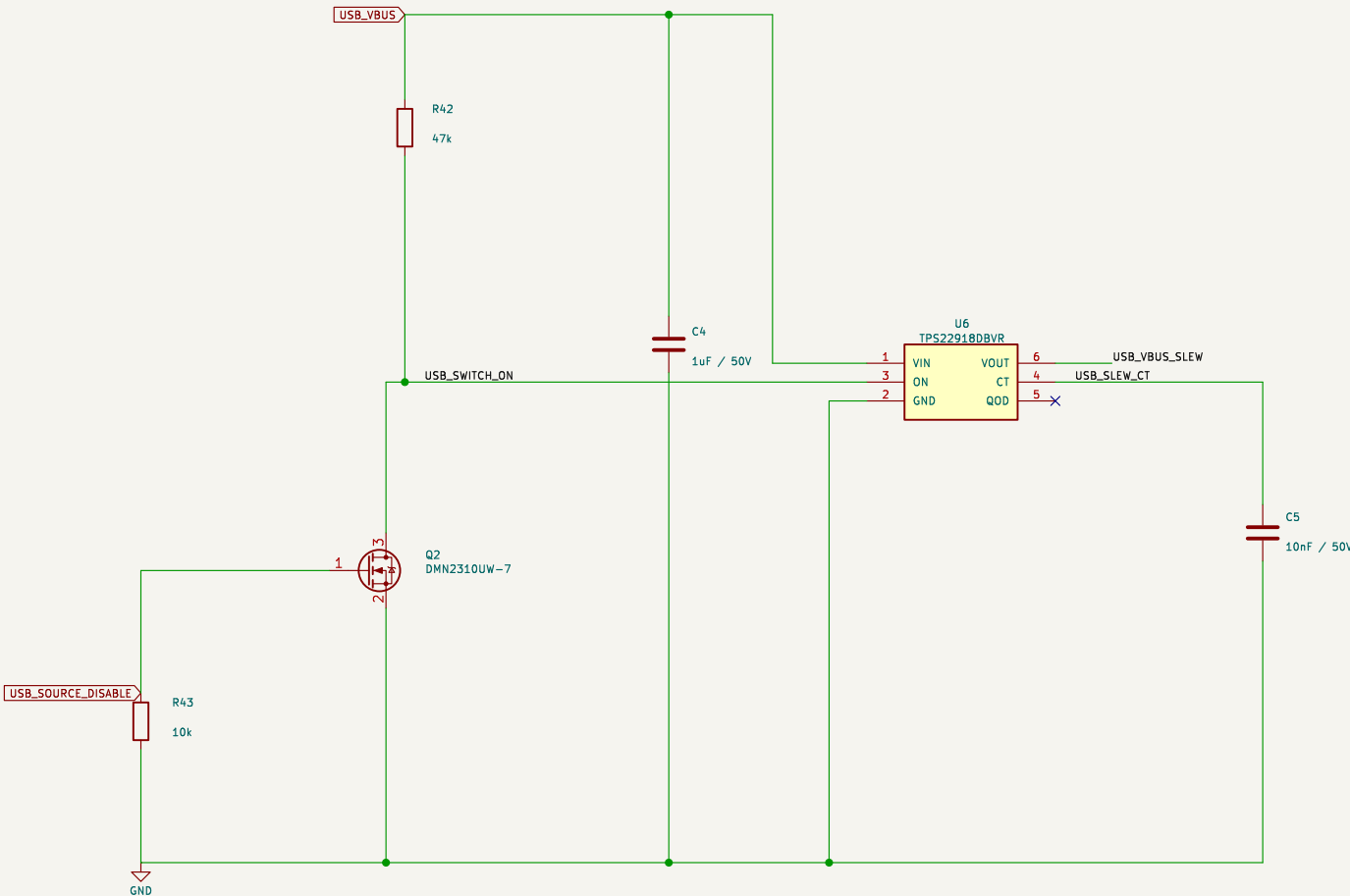
Fixed 3.3 V buck converter



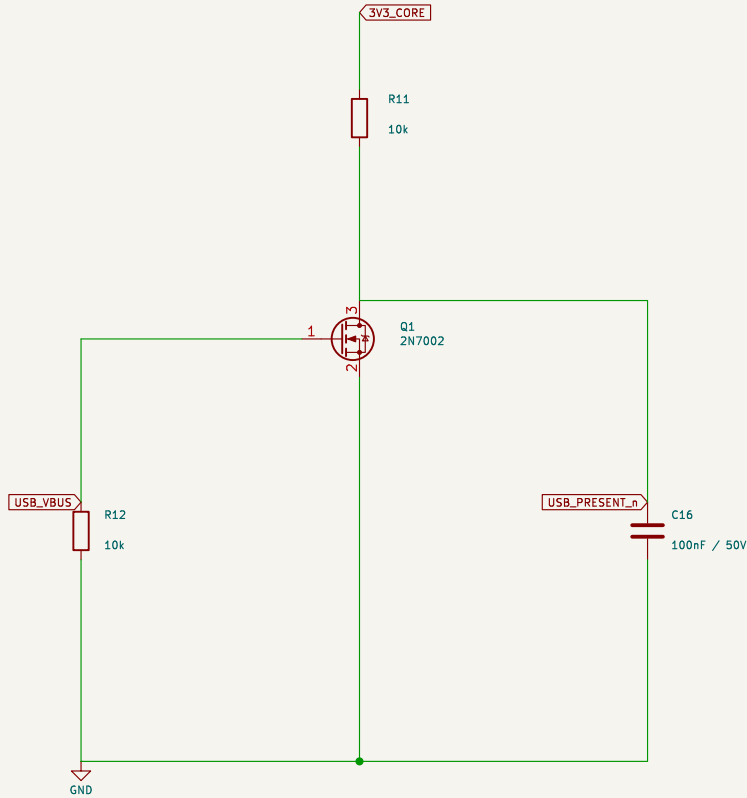
Stack 5 V health monitor



USB slew and source inhibit



USB presence detector



USB source disable is controlled by the MCU. Hardware qualification remains pending.
TPS62162 uses fixed-3.3 V option: FB grounded; VOS senses after L1.
All values, assigned footprints, component identities, and net names are preserved.

USB2 FS device:PA11/12. Separate Rd onCC1/CC2.
PA9 isUART1; disable native hardware VBUS sense.
PE10 USB_PRESENT_n must gate attach/soft-disconnect.
H743 integrates termination; R7/R8 areOR tuning links.

TPS62162 fixed3V3:FB andEP toGND; VOS senses output atC2.
PG unused; high-Z off states do not guarantee reset.
1A-rated stage600mA onboard planning allocation only.
3.3uH +22uF output plus distributed MCU bypass.
U13 mux-replaces D1/D2; MODE=PR1=GND.
Highest input selected; ST unused. No seamless-switch guarantee.
USB Inrush/dropout/current and thermal qualification still open.

USB bootstrapON. Disable only after STACK_GOOD = MOTOR_OK
and fresh motor exchange. Flight targets leave disableLOW,
USB <=100mA before configuration, then assigned current;
suspend<=2.5mA. Buck startup requires fresh Inrush check.

Stack health4.40V rise/4.34V fall; source proof still needs live motor exchange.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /power/
File: power.kicad_sch

Title: USB-C / BUCK / SOURCE CONTROL

Size: A2 Date: 2026-10-03

KiCad E.D.A. 9.0.2

Rev: R3-S6-P6
Id: 2/7

IMU / MOTOR LINK / EXTERNAL I2C

IMU filtered supply and bypass

ICM42688P:SPI1 PA5/6/7,CS PA4,INT PC4.
Orientation/vehicle-forward require target and physical tests.
Quiet filtered sensor rail: common continuous ground.

External I2C pull-ups

ExternalI2C1 PB8/PB9 +PC13 1RQ:4.7k pulls to3V3.
Corrected earlier narrative PB6/PB7: electrical map unchanged.
External sensor supply budget is not an extra regulator rating.

IMU_3V3

VDDA

3V3_CORE

Motor feedback: Schmitt buffer + pull-downs

UART8 PE1TX/PE0RX; RUN PC5; MOTOR_OK PB2.
RUN HIGH only permits an independent lower hardware gate.
MOTOR_OK HIGH=healthy; LOW/open/unpowered=unsafe.
U10 protects received UART/OK when host rail is off.
Lower board needs mirrored receivers, RUN pull-down,
VM/reset/3V3 interlocks and command-age/watchdog timeout.

Host link series damping / RUN default low

Flight targets leaveRUN LOW. Custom gimbal protocol is not supplied
by compiling Betaflight/Rotorflight/INAV/ArduPilot.
CRC/sequence/age/arm state and deliberate re-arm mandatory.
Old motor board stays paused and is not directly compatible.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

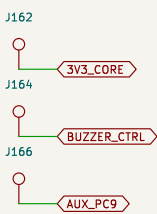
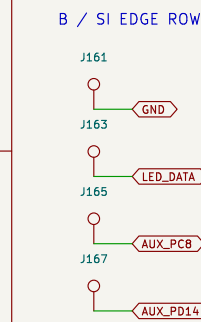
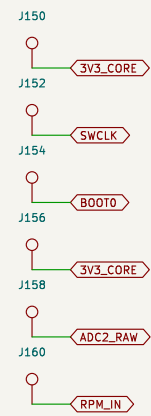
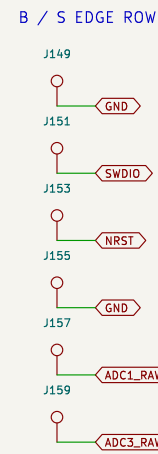
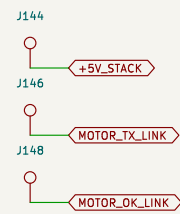
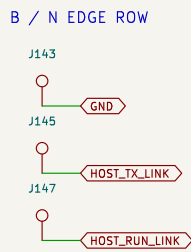
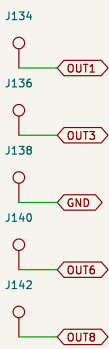
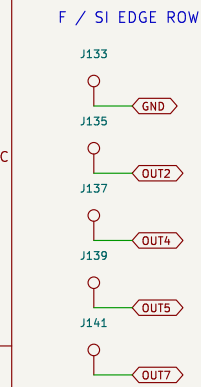
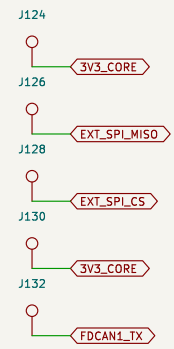
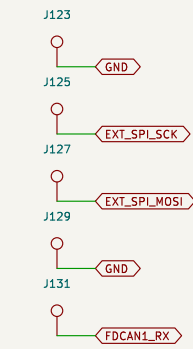
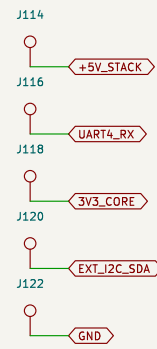
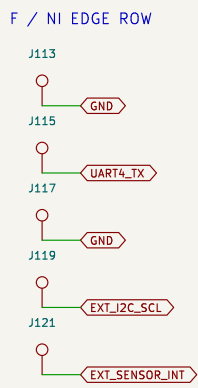
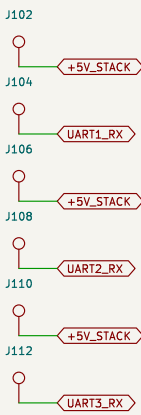
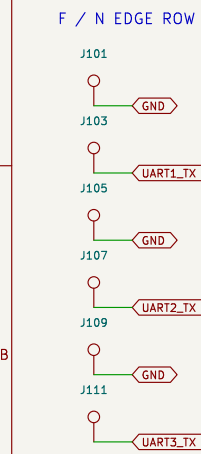
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Title: IMU / MOTOR LINK / I2C
Size: A3 Date: 2026-10-03 Rev: R3-S6-P6
KiCad E.D.A. 9.0.2 Id: 3/7

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Title: IMU / MOTOR LINK / I2C		
Size: A3	Date: 2026-10-03	Rev: R3-S6-P6
KiCad E.D.A. 9.0.2		Id: 3/7

Sheet: /link/		
File: peripherals.kicad_sch		
Title: IMU / MOTOR LINK / I2C		
Size: A3	Date: 2026-10-03	Rev: R3-S6-P6
KiCad E.D.A. 9.0.2		Id: 3/7

STorM32 CONTROLLER R3-S | TWO-EDGE IO PADS / ROLE ALTERNATIVES

Routed six-layer prototype CAD | Hardware validation pending



Raw3.3V IO or externally conditioned ADC only.
No off-state drive without external isolation.
OUT pads are signals; servo/BEC/LED/buzzer power separate.
CAN requires transceiver. SPI3 separate from NAND.

Regulated 5V input only, no direct battery.
Firmware compile checks are not bench/flight qualification.
FLASH_CS service access is internal TP1.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /pads/
File: edge-pads.kicad_sch

Title: TWO-EDGE IO PADS / ROLE ALTERNATIVES

Size: A3	Date: 2026-10-03
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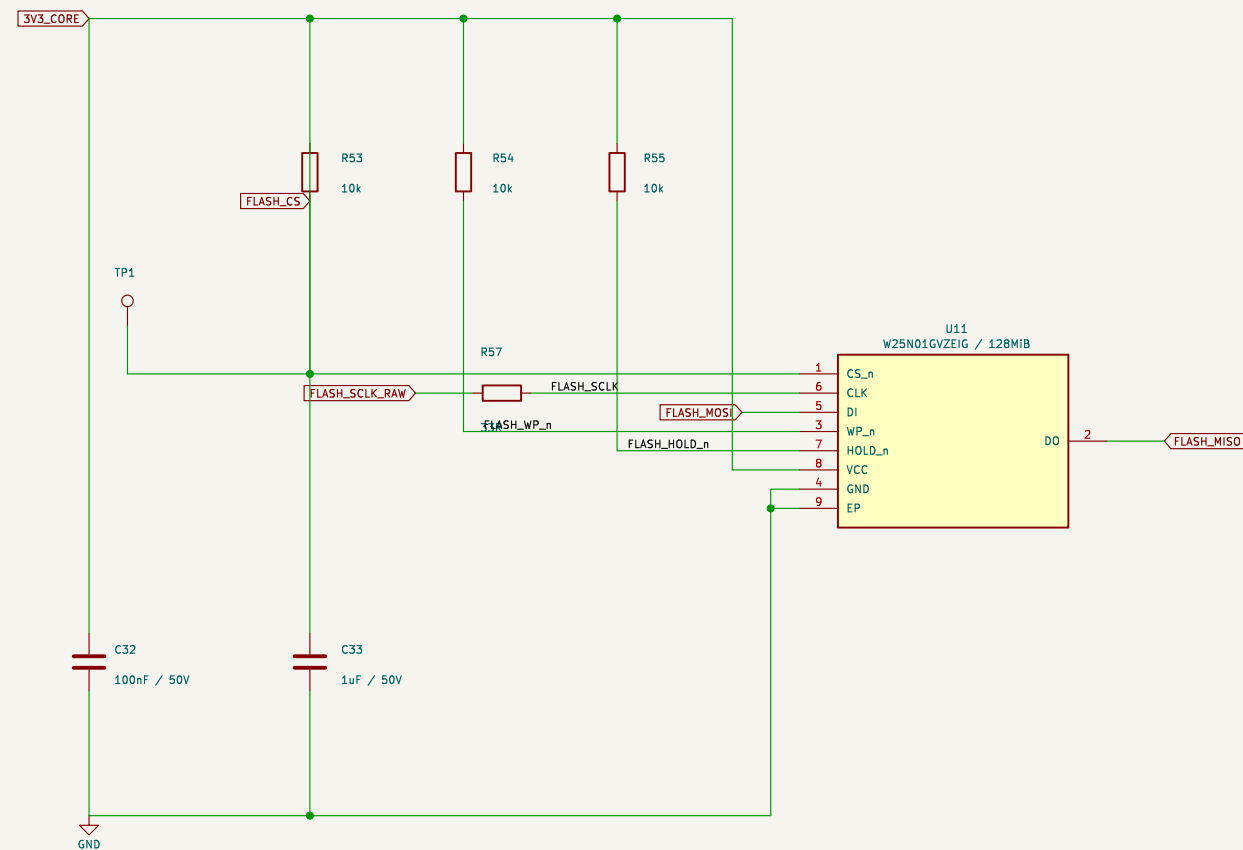
Rev: R3-S6-P6

KiCad E.D.A. 9.0.2

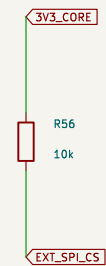
Id: 4/7

SPI NAND / PULL-UPS / LOCAL BYPASS

W25N01GVZEIG / 128 MiB



External SPI CS pull-up



WP# / HOLD# biased high. Series clock resistor at source path.
TP1 remains on FLASH_CS; all original net names retained.

PB13 SCK via33R;PB14 MISO;PB15 MOSI;PB12 CS.
WP/HOLD pulledHIGH; standard1-bitSPI. ExternalSPI separate.
R56 pulls independent externalSPI3 CS high.
35mA maximum NAND activity is inside shared core budget.
100nF+1uF bypass; no exposedPCB vias beneathEP.

Bad blocks/reserves reduce usable capacity; not MCU code expansion.
 Waitsms after stableV3, then pollBUSY; MCU reset is not NAND reset.
 HandleECC, bad blocks, page order, status errors and crash-safe metadata.
 No write-survival guarantee from supervisor or bulk capacitance.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /storage/
File: storage.kicad_sch

Title: 128 MiB NAND / DEDICATED SPI2

Size: A3	Date: 2026-10-03
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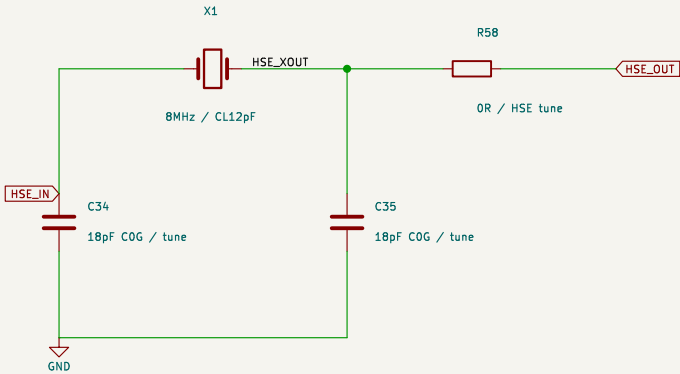
Rev: R3-S6-P6

Size: A3	
KiCad E.D.A. 9.0.2	

Id: 5/7

CLOCK / RESET / BOOT

8 MHz crystal network

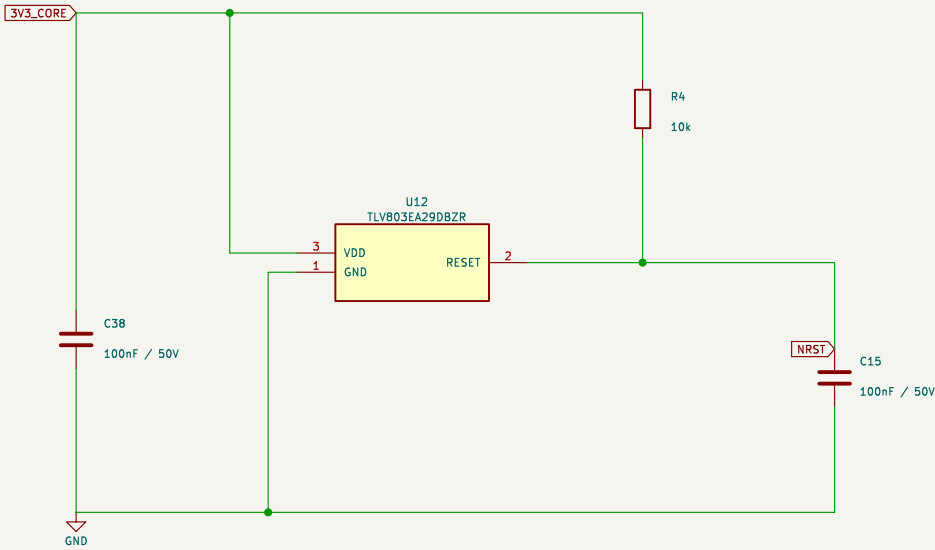


Crystal load capacitors: tune from oscillator validation
Preserved values: 18 pF COG, 0 ohm series tuning link

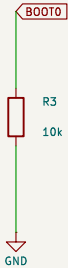
PH0pin12/PH1pin13: TXC A308000003:8MHz,CL12pF.
2x18pF assumes3pF effective stray; tune on hardware.
ESR3000Ohm,C0<=3pF preliminary gmcrit0.682mA/V.
R58 drive-tuning position. Exact-lot/startup/drive checks remain.

InternalLDO mode; correctVOS/flash wait states/clocks, HSE_VALUE=8000000.
480MHz requires compatible silicon/temperature. BOOT_ADD0/1 recovery.
SWD primary; verifyUSBDFU and cold power cycle.
RUN PC5/source-disable PE12 selected away from ROM output drive.

2.9 V reset supervisor + pull-up / filter



BOOT0 default low



EA29-2.93V open-drain reset,200ms release.
NRST shared withSWD/reset pad;BOOT0 defaults10kLOW.
EA29 avoids EA30 release-margin overlap with buck tolerance.
Interrupted NAND writes can still fail. Test brownout ramps.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /clock/
File: clock-reset.kicad_sch

Title: 8MHz HSE / RESET / RECOVERY

Size: A3 Date: 2026-10-03

KiCad E.D.A. 9.0.2

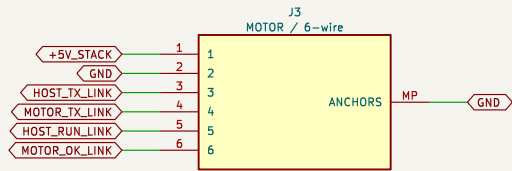
Rev: R3-S6-P6

Id: 6/7

STorM32 CONTROLLER R3-S | SIDE SOCKETS / SHARED EDGE RESOURCES

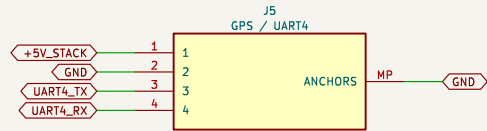
Routed six-layer prototype CAD | Hardware validation pending

J3 MOTOR:Lian Xin6, rearwest; custom pinout



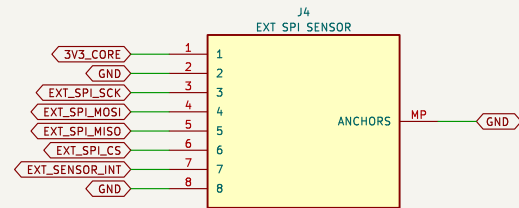
1 5V_STACK input /2GND /3hostTX /4motorTX /5RUN /6MOTOR_OK.
Duplicates existing link after damping and before receive buffer.
Use socket OR pads for one endpoint. Anchors are not extra cable returns.
Motor board needs protected inputs and default-off independent gate.
J3 carries total controller+GPS supply current; no1A per-port allocation.

J5 GPS UART4:JST4, reareast



1 5V_STACK output /2GND /3controllerTX /4controllerRX.
TX goes to moduleRX and viceversa; signals3.3V.
No5V onUSB-only power; do not connect a competing supply.
CompassI2C andPPS are not carried by this socket.
Verify module cable pinout; no universalGPS plug compatibility.

J4 EXTERNALSPI SENSOR:JST8, rearwest



1 3V3 /2GND /3SCK /4MOSI /5MISO /6CS /7IRQ /8GND.
DuplicatesSPI3 andPC13 IRQ pads. Not an independent sensor bus.
One selectedCS/IRQ owner; do not parallel push-pull outputs.
Short cable,3.3V, no off-state drive. No qualified spare power allowance.
External sensor driver/clock/damping require target and bench tests.

Friction retention; correct crimps and independent strain relief.
J3 usesZYA1001H mate; mated envelope/cross-mating unqualified.
J4/J5 plainSHR-08V-S/SHR-04V-S; use rated wire/contact combination.
2.9mm headers;JST mated2.95mm, Plan4mm rear airgap plus opposite parts.
USBovermold/harness/stack checks remain. Power off before mating.

ROUTED PROTOTYPE CAD; HARDWARE VALIDATION PENDING

Sheet: /sockets/
File: side-sockets.kicad_sch

Title: SIDE SOCKETS / SHARED EDGE RESOURCES

Size: A3	Date: 2026-10-03	Rev: R3-S6-P6
KiCad E.D.A. 9.0.2		Id: 7/7